

The Next Generation Multimodal Conversational Search and Recommendation

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ABSTRACT

The world has become multimodal. In addition to text, we have been sharing a huge amount of multimedia information in the form of images and videos on the Internet. The wide spread use of smart mobile devices has also changed the way we interact with the Internet. It is now natural for us to capture images and videos freely and use as part of a query, in addition to the traditional text and voices. These, along with the rapid advancements in multimedia, natural language processing, information retrieval, and conversation technologies, mean that it is time for us to explore multimodal conversation and its roles in search and recommendation. Multimodal conversation has the potential to help us to uncover and digest the huge amount of multimedia information and knowledge hidden within many systems. It also enables a natural 2-way interactions between humans and machines, with mutual benefits in enriching their respective knowledge. Finally, it opens up the possibilities of disrupting many existing applications and launching new innovative applications. This panel is timely and aims to explore this emerging trend, and discuss its potential benefits and pitfalls to society. The panel will also explore the limitations of current technologies and highlight future research directions towards developing a multimedia conversational system.

CCS CONCEPTS

• **Information systems** → **Multimedia and multimodal retrieval**; *Question answering*; • **Computing methodologies** → **Discourse, dialogue and pragmatics**.

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1 THE MULTIMODAL CONVERSATIONAL FUTURE

Conversational systems have been gaining popularity in recent years following many successful commercial rollouts including Amazon's Alexa, Apple's Siri and many commercial chat systems such as the Microsoft's Xiaoice etc. Much of the success comes from the maturity of multimedia content processing technologies, the availability of huge amount of multimodal data and chat sequences from various platforms, and the vast improvements in computation power and platforms. Meanwhile, the popularity of smart mobile devices along with the ease of capturing images and videos, permits users to easily and naturally pose multimodal queries and participate in multimodal conversations. This has important implications on the way we interact, retrieve and digest information from the systems. There is a strong need for new paradigms for multimodal interactions and conversations, and new approaches to carrying out searches and performing recommendation. Moreover, it offers a two-way interactive environment between users and system to learn from each other, and to enrich various types of knowledge, from domain to systems and users. Going forward, it opens up possibilities of disrupting existing applications, and conceiving new innovative applications.

To achieve the goal of developing such multimodal conversation systems that benefit both users and system, we need to tackle a range of challenges. The first challenge is on how to extend existing research to incorporate multimodal data. This would impact existing research on basic conversational models; multi-modal and cross model task oriented systems; robust and diversify conversation models; and the development of conversations systems for new tasks with low resources, etc. The second challenge is on new paradigm for conversations involving multiple modalities between humans and systems. What modalities are more natural for different types of queries and users, and when to offer more information than requested in order to provide a better context and hence shorten the process towards a successful conversation. The third challenge is how to leverage the chat sequences to extract knowledge of domain, users and system. Such knowledge can be used to enrich the system, so that it can offer more up-to-date knowledge and better understand the users. It is also beneficial to "educate" the users to be a better and more knowledgeable user in future. The fourth challenge is the same for all new research, which is the lack of large scale datasets from which to train the model. In particular, we

need datasets that are diverse and relatively complete in order to simulate real users, and towards developing robust systems. Finally, the multimodal conversation research will spur exploration to new applications, and offer a new approach to extend and possibly disrupt existing applications. As we move into the realm of real-world applications, many practical and robustness problems will arise, offering new challenges in research.

2 THE PANEL

It is therefore timely for us to organize this panel to explore multimodal conversation and its impact on search and recommendation. The panel will explore the trends, key challenges and research, as well as emerging applications in multimedia conversation search and recommendation. The topics to be discussed include, but not limited to, the followings.

- Roles and impact of multimodality in conversation
- New paradigms in multimodal conversation
- Key research trends in multimodal conversational search and recommendation (MCSR)
- Learning and knowledge acquisition through conversation
- Emerging applications of MCSR in Ecommerce, healthcare, legal and others
- Limitations, datasets and future trends
- Implications of new technologies to society

We have formed a panel comprising experts from both academia and industry, as follows.

- Moderator: Alan Smeaton (Dublin City University)
- Panelists:
 - Alex Hauptman (Carnegie Mellon University)
 - Tat-Seng Chua (National University of Singapore)
 - Claudia Hauff (Delft University of Technology)
 - Tao Mei (JD.COM)
 - Verena Rieser (Alana.AI)

The panel will be organized as an interactive discussion in different rounds, guided by the Moderator. Each panelist will contribute a short statement of his/her position which will be discussed within the panel and with the wider audience.

This panel is organized to complement the Workshop on Multimodal Conversational AI to be held as part of ACM Multimedia 2021. It is hoped that this panel would draw curiosity and interests of our community towards this topic, while many of the topics discussed during this panel would follow naturally into the Workshop to be held at the end of this conference.

3 THE PANELISTS

Prof Alan Smeaton is Professor of Computing at Dublin City University where he has previously been Head of School and Executive Dean of Faculty. He is a founding Director of the Insight Centre for Data Analytics at DCU and an elected member of the Royal Irish Academy and Academy Gold Medal winner in 2016. From 2001 until recently he was a coordinator of the annual TRECvid benchmarking activity for content-based video operations. His research interests are in developing computer systems which help people find information in support of human memory. Alan was Chair of SIGMM from 2017 to 2021.

Alex Hauptmann is a Research Professor at Carnegie Mellon University in the Language Technologies Institute of the School of Computer Science. He received B.A. and M.A. degrees in Psychology from Johns Hopkins University. He studied Computer Science at the Technische Universität Berlin from 1982 to 1984 and received his Ph.D. degree in Computer Science from CMU in 1991. His research interests have led him to pursue and combine several different areas: video analysis, natural language processing, speech understanding and synthesis, man-machine communication, and machine learning. He has conducted research projects on multimedia analysis and retrieval on observational video with success documented by outstanding performance in the annual NIST TRECVID video analysis and surveillance event detection evaluations.

Tat-Seng Chua is the KITHCT Chair Professor at the School of Computing, National University of Singapore (NUS). He is the Founding Dean of the School of Computing from 1998-2000; and is the co-Director of NExT, a joint research Center between NUS, Tsinghua University and Southampton University, focusing on Extreme Search. His main research interests include unstructured data analytics, multimedia information retrieval, recommendation and conversation systems. He has initiated research on multimodal conversation in search and recommendation in the domain of e-commerce and Fintech. He is particularly interested in knowledge acquisition through conversations.

Claudia Hauff is an Associate Professor at the Web Information Systems group, Delft University of Technology (TU Delft), and a computer scientist by training. She received her Ph.D. in 2010 from the University of Twente. In the past, she has worked on a variety of topics in the fields of information retrieval and data science, including query performance prediction, social search, learning to search, and information retrieval for specific user groups. Together with her team she currently focuses on the areas of collaborative search, complex search, and conversational search.

Tao Mei is a Vice President with JD.com and the Deputy Managing Director of JD AI Research. Prior to joining JD.com in 2018, he was a Senior Research Manager with Microsoft Research Asia. He has authored or co-authored over 200 publications (with 12 best paper awards) in journals and conferences. He is or has been an Editorial Board Member of leading multimedia journals, and the General/Program Chairs of premier multimedia conferences. He was elected to a Fellow of IEEE, a Fellow of IAPR, and a Distinguished Scientist of ACM, for his contributions to large-scale multimedia analysis and applications.

Verena Rieser is a co-founder of the UK company ALANA.AI and full professor in Computer Science at Heriot-Watt University in Edinburgh, where she leads research on Conversational AI at the intersection of Natural Language Processing and Machine Learning. She co-authored a total of over 100 publications in these areas. Verena was recently awarded a Leverhulme Senior Research Fellowship by the Royal Society in recognition of her work in developing multimodal conversational systems. She is a regular academic keynote speaker and represents ALANA.AI at industry and public outreach events. Alana.AI are currently seeking growth funding.